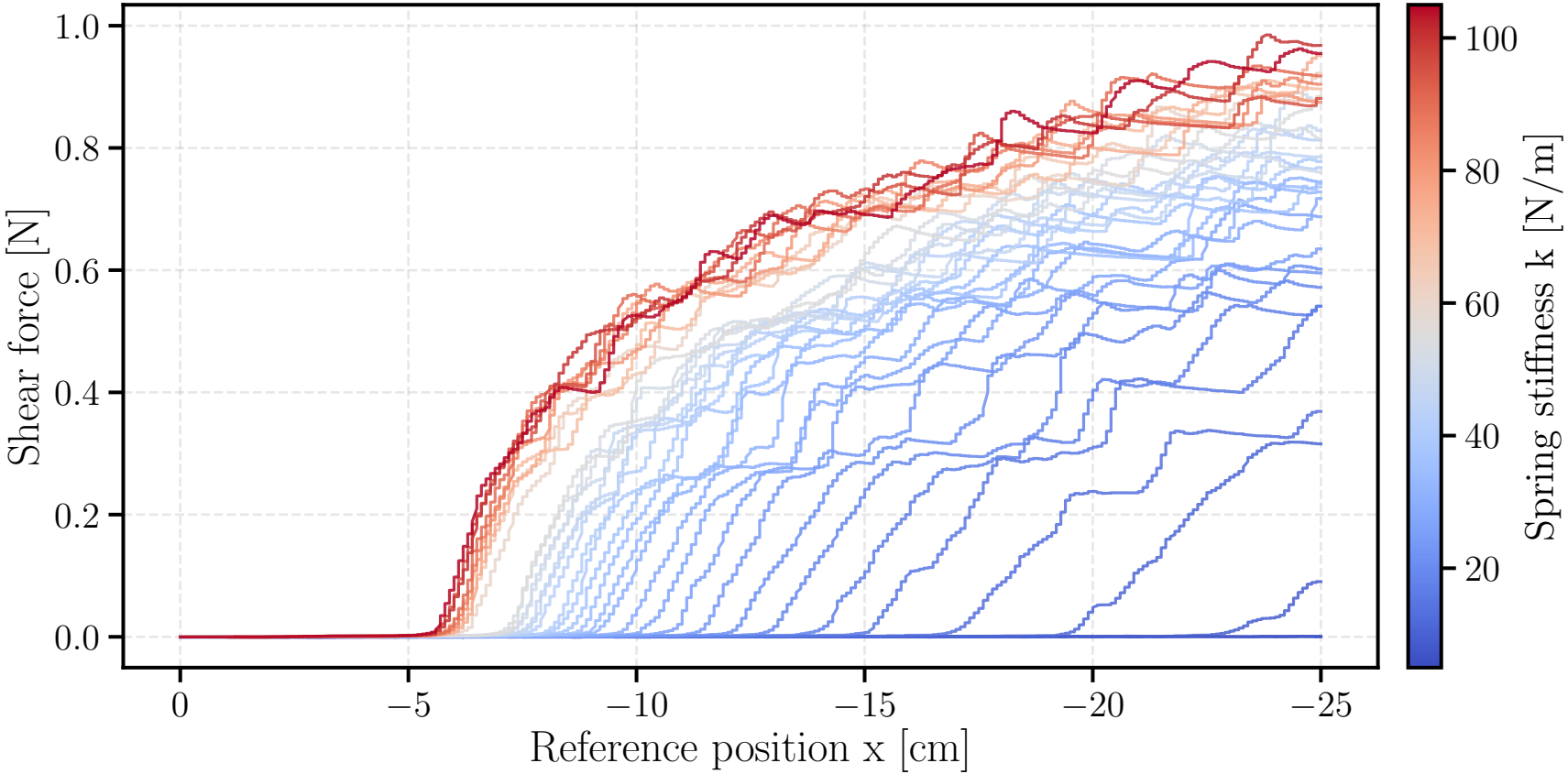

Shear force [N]
1.0
0.8
0.6
0.4
0.2
0.0
0
−5
−10
−15
−20
−25
Reference position x [cm]
Spring stiffness k [N/m]
100
80
60
40
20